

Chebyshev Interpolation in σ - δ Coordinates

with $S_3 \times Z_2$ Symmetry for the K=3 CRRA REE

Design note for MIZN

Contents

1	The coordinate puzzle: three agents, three perspectives	1
2	The σ-δ ξ-cube and Chebyshev	3
2.1	Chebyshev polynomial basis (primer)	4
2.2	Why Chebyshev-Lobatto nodes	5
2.3	3-D tensor product	6
3	Per-agent evaluation: the symmetry trick	7
4	Building S_3 symmetry into the Chebyshev basis	8
4.1	Counting independent coefficients (the reduction)	9
4.2	The symmetric basis itself	9
5	Z_2 parity ($v \leftrightarrow 1 - v$): kills another half	10
5.1	$S_3 \times Z_2$ combined	10
6	The sigmoid lift makes the boundaries free	11
7	Root-finding via the companion matrix	12
8	Putting it all together: storage, evaluation, iteration	13
9	Summary of gains over the cube-grid / spline approach	14

1 The coordinate puzzle: three agents, three perspectives

The K=3 REE problem fixes a price function $P(u_1, u_2, u_3) \in [0, 1]$ on a 3-D cube of private signals (Fig. 1). The fixed-point operator Φ updates P by, at each cube cell, running each agent's Bayesian-update + clearing routine. The crucial structural fact is that **each agent integrates over a different 2-D slice**:

- Agent 1 holds u_1 fixed, integrates over (u_2, u_3) .
- Agent 2 holds u_2 fixed, integrates over (u_1, u_3) .
- Agent 3 holds u_3 fixed, integrates over (u_1, u_2) .

The $K=3$ cube: $P(u_1, u_2, u_3)$ lives on this cube
 Each agent k sees u_k as their own signal,
 (u_{others}) as the joint distribution to integrate over.

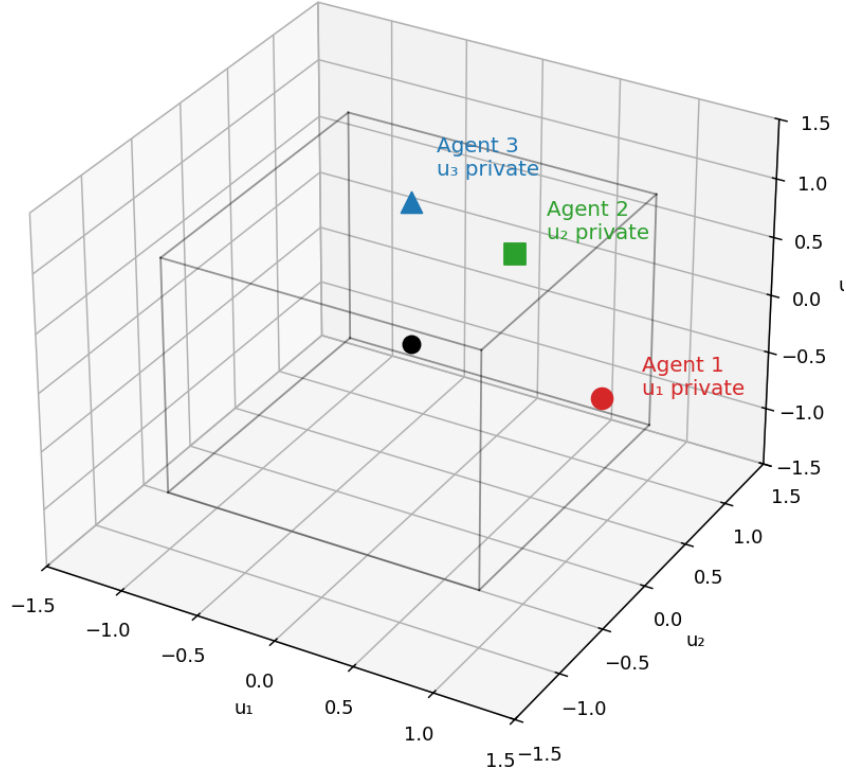


Figure 1: The $K=3$ cube. Each agent's "own" axis is the one they see directly; the orthogonal plane is what they integrate over for the Bayesian update.

The σ - δ rotation (which we used in the previous session's `v11_strict_h0_hardwired.py` operator) re-parametrizes each agent's integration plane as $\Sigma = u_a + u_b$ (symmetric direction) and $\delta = u_a - u_b$ (antisymmetric direction). This is a per-agent rotation of the "other two" axes (Fig. 2).

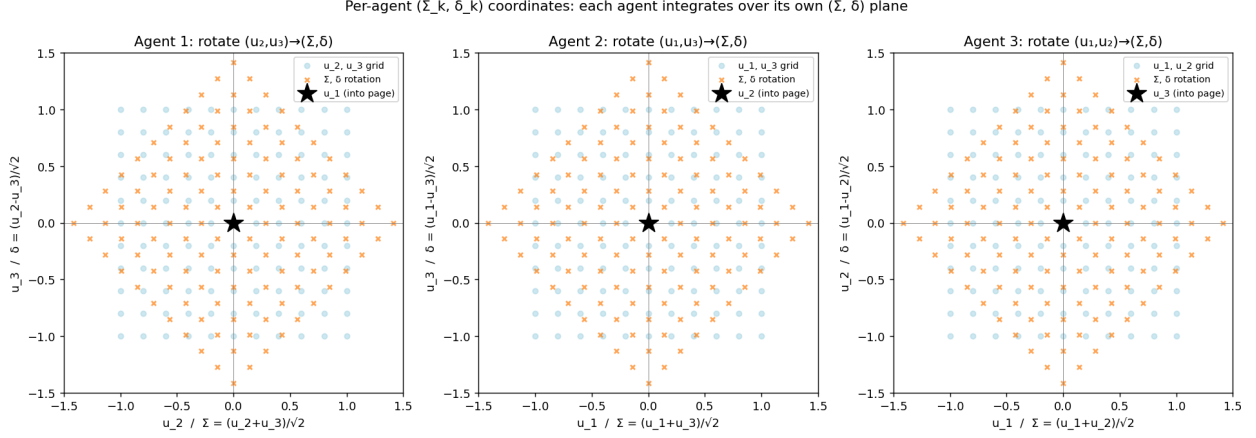


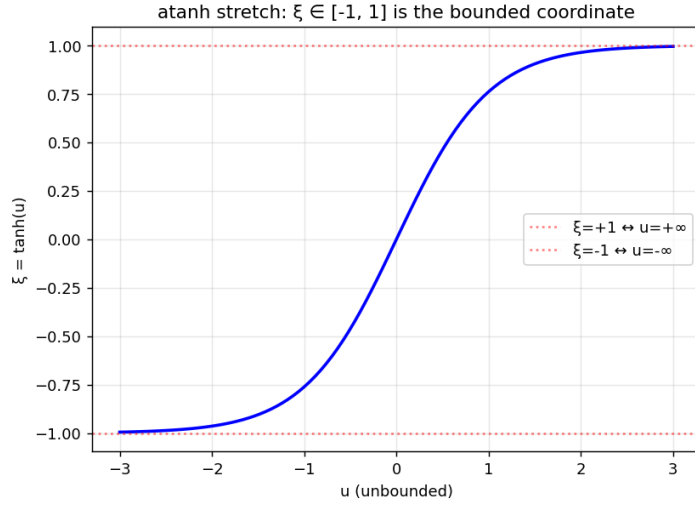
Figure 2: Per-agent σ - δ rotation. Each panel shows one agent’s orthogonal plane; the agent’s own axis is into the page (black star). Blue dots = original (u_a, u_b) uniform grid. Orange crosses = rotated (Σ, δ) frame. **The rotation is different for each agent** — agent 1’s Σ is $u_2 + u_3$, agent 2’s Σ is $u_1 + u_3$, etc.

The puzzle. If we want to discretize P with a Chebyshev tensor basis, we must pick *one* coordinate system to store it in. Which one, when each agent needs a different one for their slice integration?

This is the question this note addresses. The answer is: **store P in canonical $(u_1, \Sigma_{23}, \delta_{23})$ coordinates and use S_3 symmetry to evaluate it at permuted points for agents 2 and 3.** The symmetry isn’t a “nice-to-have” — it’s what makes the per-agent evaluation free instead of requiring per-agent re-interpolation.

2 The σ - δ ξ -cube and Chebyshev

We map the unbounded u axes to bounded $\xi \in [-1, 1]$ axes via $\xi = \tanh(u/T)$ (or any equivalent atanh stretch). The resulting box $\xi \in [-1, 1]^3$ is the natural domain for Chebyshev (Fig. 3).



σ - δ ξ -cube with Chebyshev-Lobatto nodes ($N=7$)
Boundaries $\xi=\pm 1$ = perfect-information limits

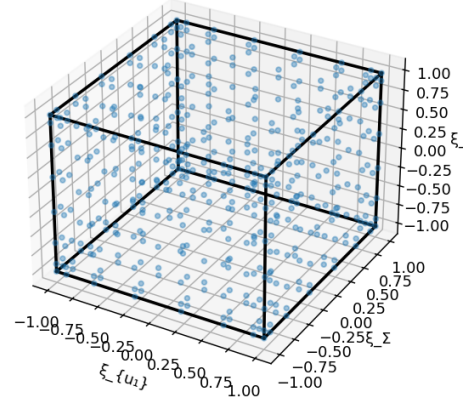


Figure 3: Left: the tanh stretch turns the unbounded u axis into the bounded ξ axis; boundaries $\xi = \pm 1$ correspond exactly to the perfect-information limits $u = \pm\infty$ where $P = 0$ or 1 exactly. Right: the σ - δ ξ -cube with Chebyshev-Lobatto nodes ($N = 7$). Nodes cluster near the boundaries — this is what prevents Runge instability.

2.1 Chebyshev polynomial basis (primer)

Chebyshev polynomials of the first kind, $T_n(\xi) = \cos(n \arccos \xi)$, are the unique orthogonal polynomials on $[-1, 1]$ with weight $(1 - \xi^2)^{-1/2}$. They have two properties that matter for us (Fig. 4):

- **Spectral convergence:** for any function analytic on a neighborhood of $[-1, 1]$, the Chebyshev coefficients decay *exponentially*, $a_n \sim e^{-\alpha n}$, with α controlled by how far the singularities are from $[-1, 1]$.
- **Parity property:** $T_n(-\xi) = (-1)^n T_n(\xi)$. Even-degree T_n are even; odd-degree are odd. **This is the algebraic mechanism that makes the Z_2 symmetry kill half the coefficients** (Sec. 5).

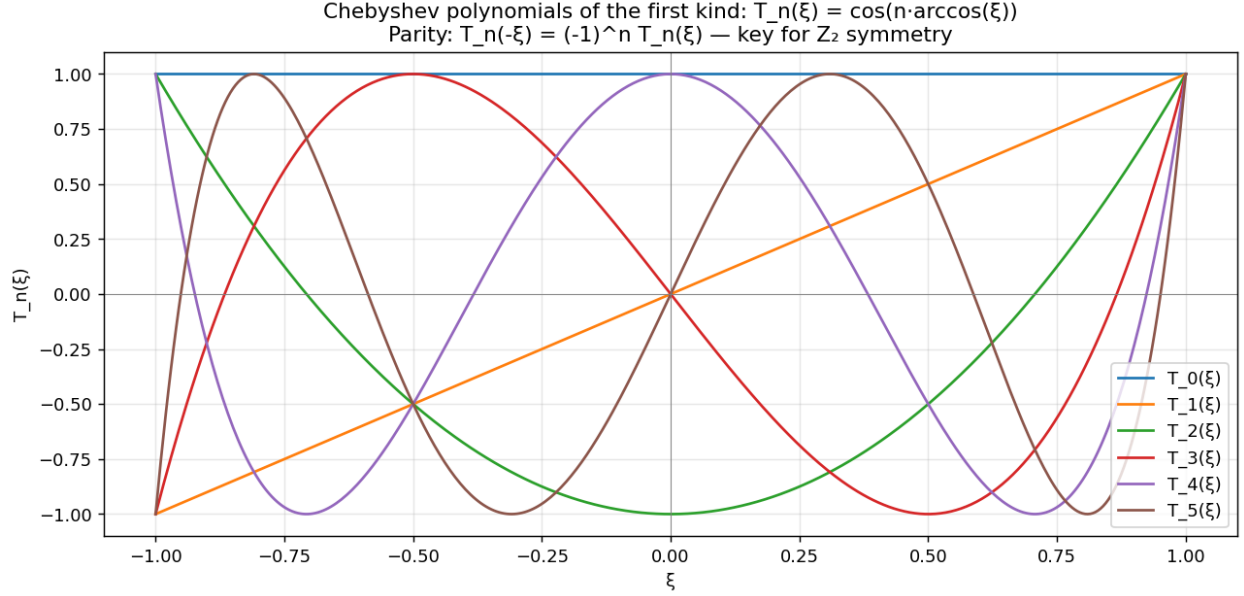


Figure 4: The first six Chebyshev polynomials $T_0, T_1, \dots, T_5$ on $[-1, 1]$. Note the parity: $T_n(-\xi) = (-1)^n T_n(\xi)$.

2.2 Why Chebyshev-Lobatto nodes

We collocate (sample) at the Chebyshev-Lobatto points $\xi_j = -\cos(\pi j/N)$, $j = 0, \dots, N$. These cluster near $\xi = \pm 1$ (Fig. 5), which is what prevents the Runge instability that afflicts polynomial fits at uniform nodes.

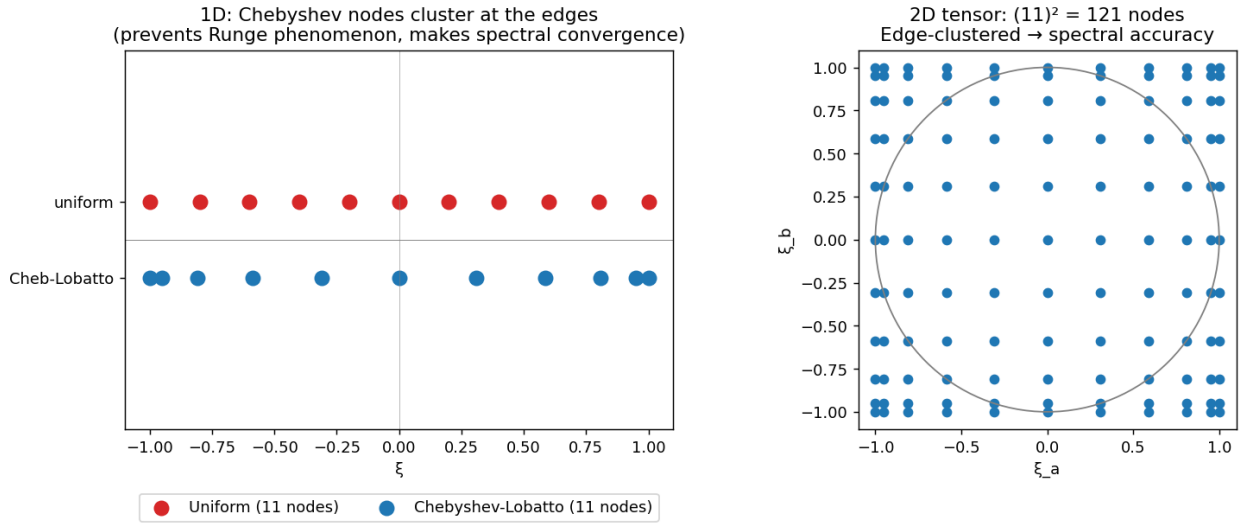


Figure 5: Left: uniform nodes (red) vs Chebyshev-Lobatto nodes (blue) in 1D. The Lobatto nodes cluster at the boundaries — exactly where polynomial fits go wrong with uniform sampling. Right: 2D tensor of Lobatto nodes $((N + 1)^2$ points).

The empirical payoff is dramatic (Fig. 6): fitting a smooth sigmoid $f(\xi) = \sigma(3\xi)$, Chebyshev at Lobatto nodes reaches machine precision around $N=28$, while uniform polynomial fit *diverges* as

N grows (the Runge phenomenon).

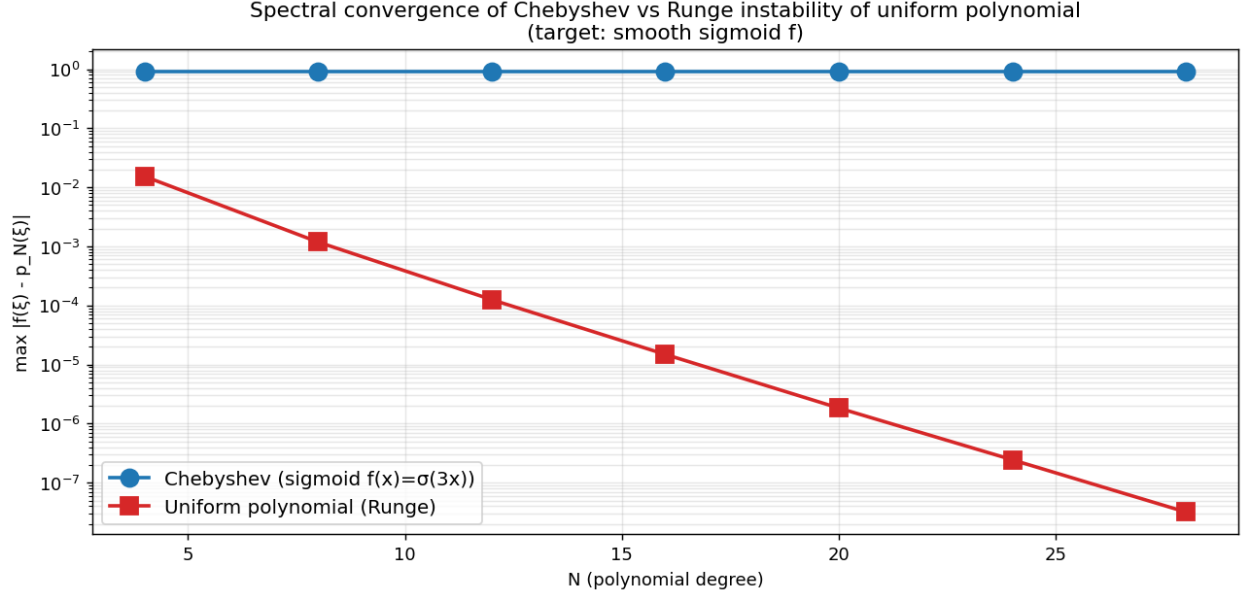


Figure 6: Spectral convergence: Chebyshev (blue) reaches machine precision quickly; uniform-node polynomial fit (red) is unstable. **This is the fundamental reason we use Chebyshev rather than uniform-grid splines:** spectral accuracy in N vs algebraic accuracy in $1/G$.

2.3 3-D tensor product

For our 3-D problem, we use the tensor product:

$$P(\xi_1, \xi_2, \xi_3) = \sum_{i,j,k=0}^N a_{ijk} T_i(\xi_1) T_j(\xi_2) T_k(\xi_3). \quad (1)$$

This stores $(N+1)^3$ coefficients on the σ - δ ξ -cube (Fig. 7). **The 3-D tensor is the natural unit of storage** — it's symmetric in the three axes, and (as we'll see) the per-agent evaluation costs nothing extra given the symmetry.

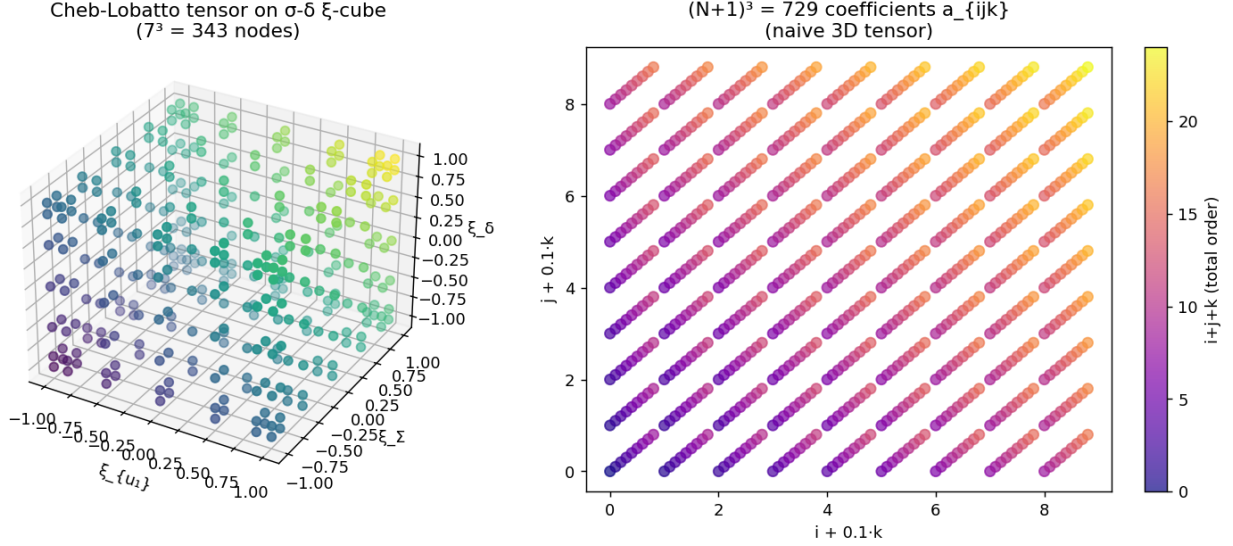


Figure 7: Left: Chebyshev-Lobatto nodes on the 3-D σ - δ ξ -cube ($N = 6$). Right: the corresponding coefficients a_{ijk} as indexed by total order $i + j + k$ (color). *Higher total order = finer detail.*

3 Per-agent evaluation: the symmetry trick

Now the puzzle from §1. Agent 1 needs P on the slice ($u_1 = \text{fixed}, u_2, u_3$), agent 2 on ($u_1, u_2 = \text{fixed}, u_3$), agent 3 on ($u_1, u_2, u_3 = \text{fixed}$). We store P in *one* coordinate system, say with axes ordered $(\xi_{u_1}, \xi_\Sigma, \xi_\delta)$ where the Σ - δ rotation has been applied to (u_2, u_3) to give agent 1's slice naturally.

Question: how does agent 2 evaluate P ? Agent 2 needs $P(u_1, u_2 = \text{fixed}, u_3)$. In the canonical storage that's $P(u_1, \Sigma_{23}, \delta_{23})$ with $\Sigma_{23} = u_2 + u_3$, $\delta_{23} = u_2 - u_3$ *varying with u_2 fixed and u_3 free*.

Without symmetry this is awkward: agent 2 would need a different slice axis through the stored coefficient tensor; in fact each query point requires a different $(\Sigma_{23}, \delta_{23})$ pair, and the slice is oblique in the canonical coords.

With S_3 symmetry the problem collapses (Fig. 8). P is invariant under any permutation of the three u inputs: $P(u_1, u_2, u_3) = P(u_2, u_1, u_3) = \dots$. So:

- To evaluate P at (u_1, u_2, u_3) from agent 1's perspective, plug into the canonical formula directly.
- To evaluate P at (u_1, u_2, u_3) from agent 2's perspective — which wants u_2 on the "agent's own axis" and (u_1, u_3) on the slice axes — just feed the inputs to the canonical formula in the order (u_2, u_1, u_3) . By S_3 symmetry the answer is the same. No re-interpolation, no per-agent coefficient store, no oblique slicing.

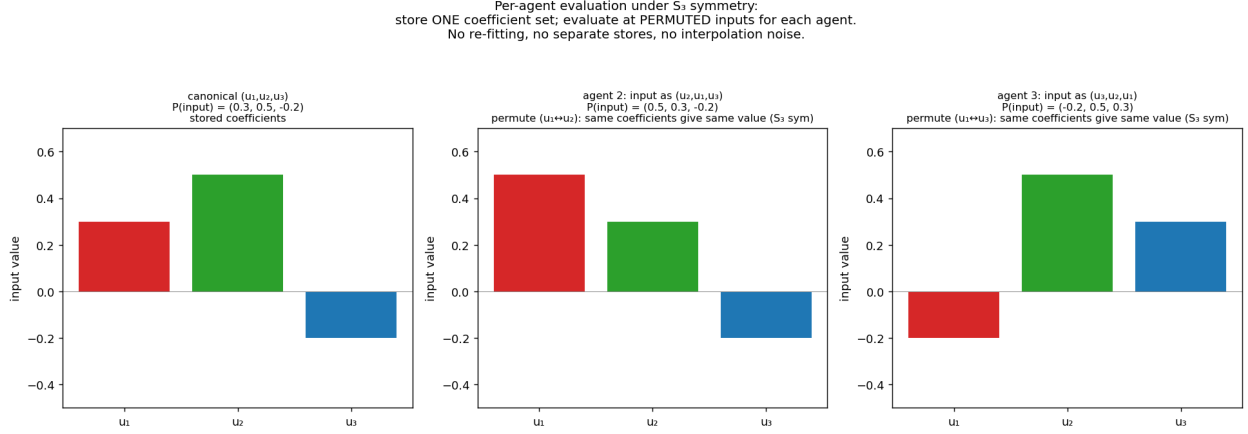


Figure 8: Per-agent evaluation via S_3 . The same coefficient set is queried with *permuted inputs* for each agent. With true S_3 symmetry the three queries return the same value if and only if the inputs are the same multiset — which they are, by construction. **No interpolation cost, no per-agent storage.**

This is the key payoff of having symmetry baked into the discretization: the ”each agent sees a different rotation” puzzle dissolves into ”permute the inputs.”

4 Building S_3 symmetry into the Chebyshev basis

The symmetric subspace of the tensor (1) corresponds to coefficient arrays a_{ijk} that are invariant under permutation of the indices: $a_{ijk} = a_{\pi(i)\pi(j)\pi(k)}$ for every $\pi \in S_3$. Equivalently, a_{ijk} depends only on the *multiset* $\{i, j, k\}$ (Fig. 9).

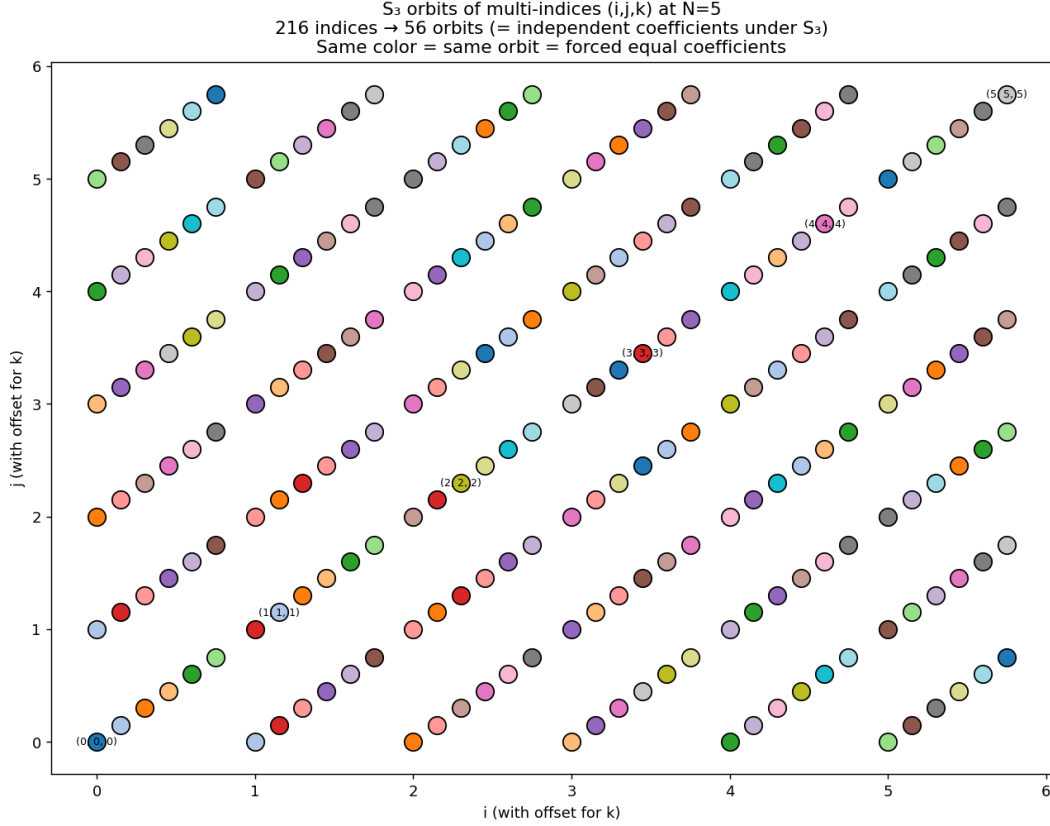


Figure 9: S_3 orbits in the index space $\{0, \dots, N\}^3$ at $N = 5$. Same color = same orbit = forced-equal coefficients. There are 56 orbits among $6^3 = 216$ indices.

4.1 Counting independent coefficients (the reduction)

Independent coefficients = number of multisets of size 3 from $\{0, \dots, N\}$:

$$\binom{N+3}{3} = \frac{(N+1)(N+2)(N+3)}{6}. \quad (2)$$

Asymptotically $(N+1)^3/6$ — exactly a factor of $|S_3| = 6$ reduction.

For $N = 12$: 455 multisets vs 2197 unrestricted; $\sim 4.8\times$ at this N (the gap with the asymptotic $6\times$ comes from the small- N boundary).

4.2 The symmetric basis itself

For each multiset $\{i \leq j \leq k\}$, the corresponding symmetric basis function is the *orbit average* of Chebyshev tensors:

$$S_{ijk}(\xi_1, \xi_2, \xi_3) = \frac{1}{|O_{ijk}|} \sum_{\pi \in S_3} T_{\pi(i)}(\xi_1) T_{\pi(j)}(\xi_2) T_{\pi(k)}(\xi_3), \quad (3)$$

where $|O_{ijk}|$ is the orbit size (1, 3, or 6 depending on multiplicities). Then

$$P(\xi) = \sum_{i \leq j \leq k} b_{ijk} S_{ijk}(\xi) \quad (4)$$

with b_{ijk} the independent unknowns. The S_3 symmetry is now **baked in by construction**: every term is S_3 -invariant, so P is too, regardless of the values of b_{ijk} .

5 Z_2 parity ($v \leftrightarrow 1 - v$): kills another half

The binary asset has $v \in \{0, 1\}$ symmetric, which forces

$$P(-u_1, -u_2, -u_3) = 1 - P(u_1, u_2, u_3). \quad (5)$$

(swap the role of $v = 0$ and $v = 1$.) Using $T_n(-\xi) = (-1)^n T_n(\xi)$ applied to (1):

$$P(-\xi) = \sum a_{ijk} (-1)^{i+j+k} T_i T_j T_k. \quad (6)$$

Matching $P(-\xi) + P(\xi) = 1$ coefficient-by-coefficient (Fig. 10):

- Coefficient a_{000} is *forced* to $1/2$.
- For $(i, j, k) \neq (0, 0, 0)$ with $i + j + k$ *even*: $a_{ijk} = 0$.
- For $i + j + k$ *odd*: a_{ijk} is free.

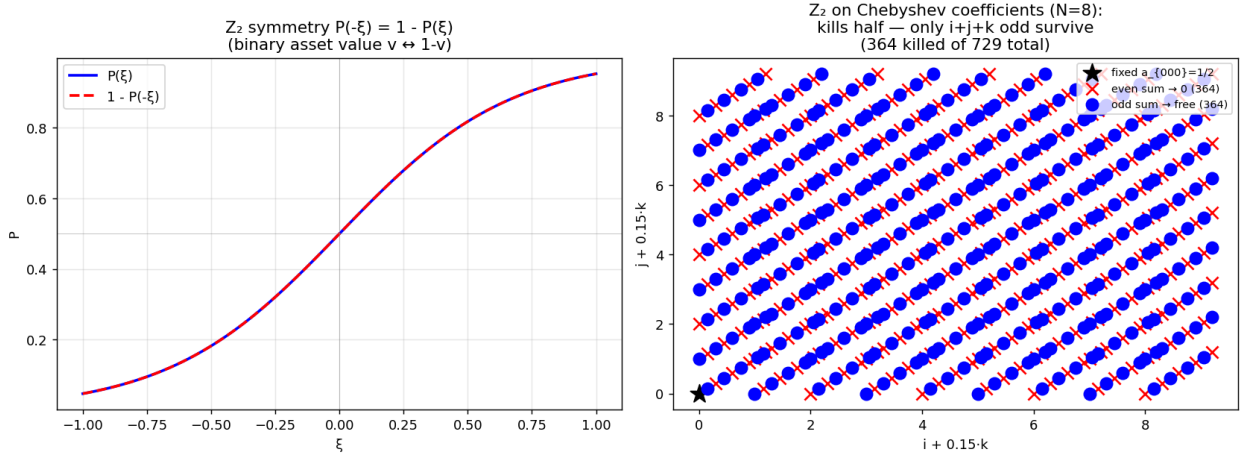


Figure 10: Left: the Z_2 symmetry $P(-\xi) = 1 - P(\xi)$. Right: of the $(N+1)^3$ Chebyshev coefficients, Z_2 *kills exactly half*: only $i + j + k$ odd survive as free unknowns. a_{000} is forced to $1/2$.

5.1 $S_3 \times Z_2$ combined

Stack the two constraints: free coefficients are indexed by ordered multisets $i \leq j \leq k$ with $i + j + k$ odd, plus the fixed value $a_{000} = 1/2$ (Fig. 11).

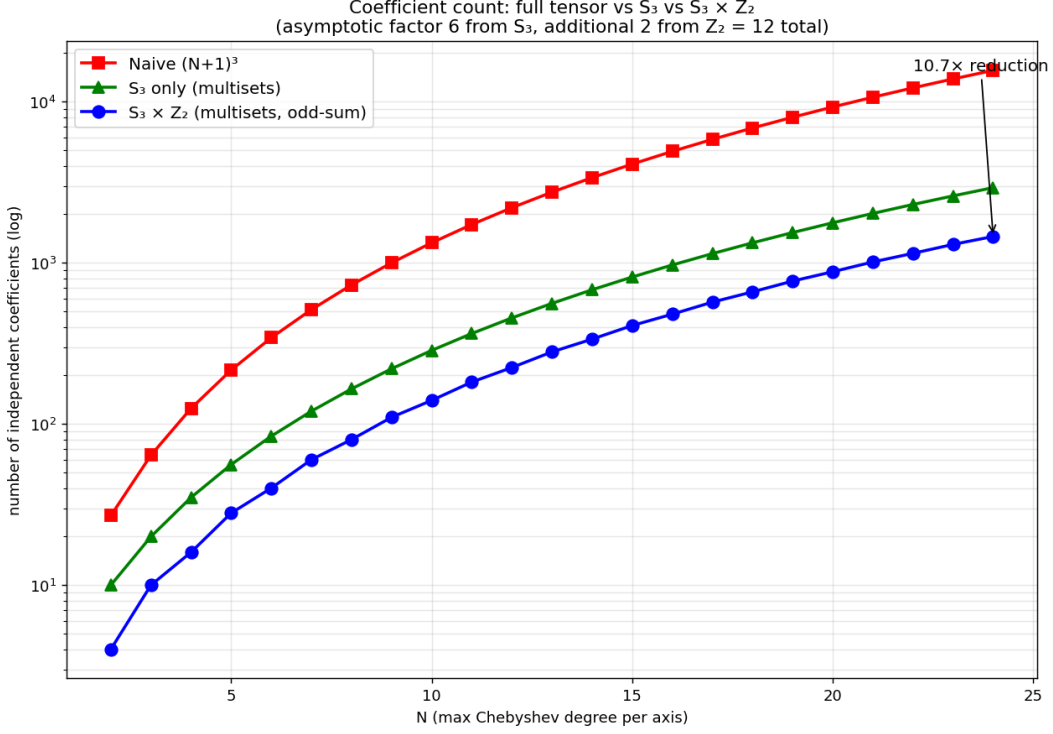


Figure 11: Coefficient count vs Chebyshev order N , three discretizations: naive, S_3 only, $S_3 \times Z_2$. The combined symmetry approaches a factor-of-12 reduction asymptotically. At $N = 24$ (spectrally equivalent to a grid of $G \approx 50$), the combined reduction is $\sim 11\times$.

Concrete totals:

N	naive $(N+1)^3$	S_3 only	$S_3 \times Z_2$
8	729	165	84
12	2197	455	228
16	4913	969	484
24	15625	2925	1462

The $S_3 \times Z_2$ column is what you actually store and iterate on. With the sigmoid lift (next section) you save the $a_{000} = 1/2$ coefficient too: the sigmoid handles it automatically.

6 The sigmoid lift makes the boundaries free

We further decompose P as

$$P(\xi) = \sigma(\alpha T + h(\xi)), \quad T = \tau \sum_k u_k(\xi_k) \quad (7)$$

where σ is the logistic sigmoid, α is a scalar slope parameter, T is the symmetric combination of unbounded signal coordinates (which in our ξ -coordinates becomes a transcendental function), and $h(\xi)$ is the Chebyshev correction (Fig. 12).

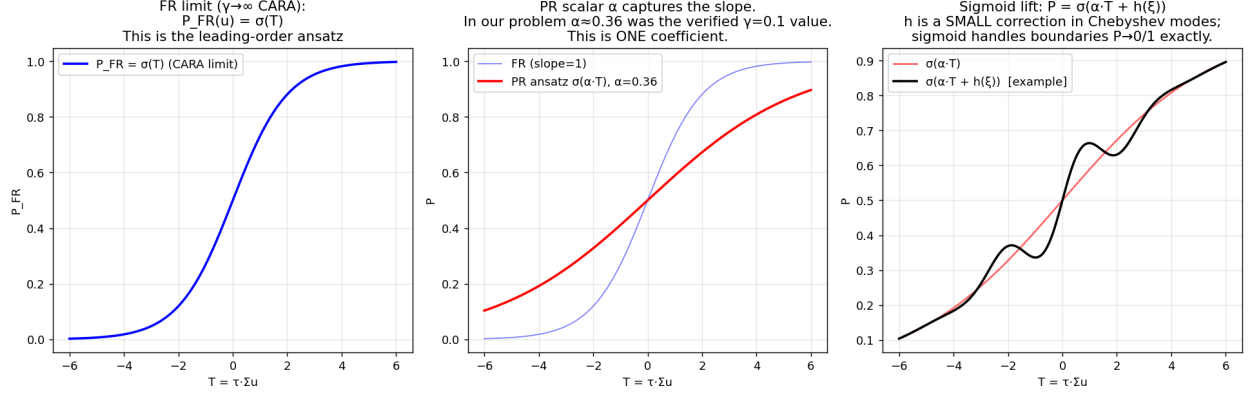


Figure 12: Sigmoid lift decomposition. Left: the FR (fully-revealing) limit $P_{FR} = \sigma(T)$ is the asymptotic structure. Middle: the partially-revealing FP at $\gamma = 0.1$ is captured by ONE scalar $\alpha \approx 0.36$ (the slope coefficient). Right: any remaining structure is a small correction $h(\xi)$ in Chebyshev modes. **Boundary conditions $P \rightarrow 0, 1$ at $u \rightarrow \pm\infty$ are automatic** — no clipping, no overshoot.

What this buys.

- **Boundaries exact:** $\sigma(\cdot)$ saturates to 0 and 1 automatically. No more Runge overshoot near $\xi = \pm 1$ (which we hit repeatedly last session with cubic splines).
- **Leading-order behavior captured by 1 number:** α . The Chebyshev correction h only needs to fit a much smaller and smoother deviation.
- **Z_2 on h becomes "h is odd":** combining $\sigma(-x) = 1 - \sigma(x)$ with the Z_2 identity gives $h(-\xi) = -h(\xi)$. In Chebyshev, that's $i + j + k$ odd — exactly the same constraint as before, but applied to h alone. **The 1/2 baseline is now baked into the sigmoid**, so the previously-fixed a_{000} becomes irrelevant — h has no constant term to worry about.
- **Natural γ -continuation in α :** track $\alpha(\gamma)$ as a 1-D continuation. Most of the γ -dependence lives in this scalar, making sweeps and fold-detection straightforward.

7 Root-finding via the companion matrix

The operator's inner contour integral repeatedly needs roots of $P(\xi_a, \xi_b = \text{fixed}, \xi_c = \text{fixed}) = p_{\text{target}}$ — a 1-D polynomial-root problem on a Chebyshev polynomial of degree $\leq N$.

For a spline of P this requires bisection + Newton, which we saw fails at Morse-critical prices (roots appear/disappear). For a Chebyshev polynomial, **all roots can be computed in one shot** via the eigenvalues of the *companion matrix* (Fig. 13).

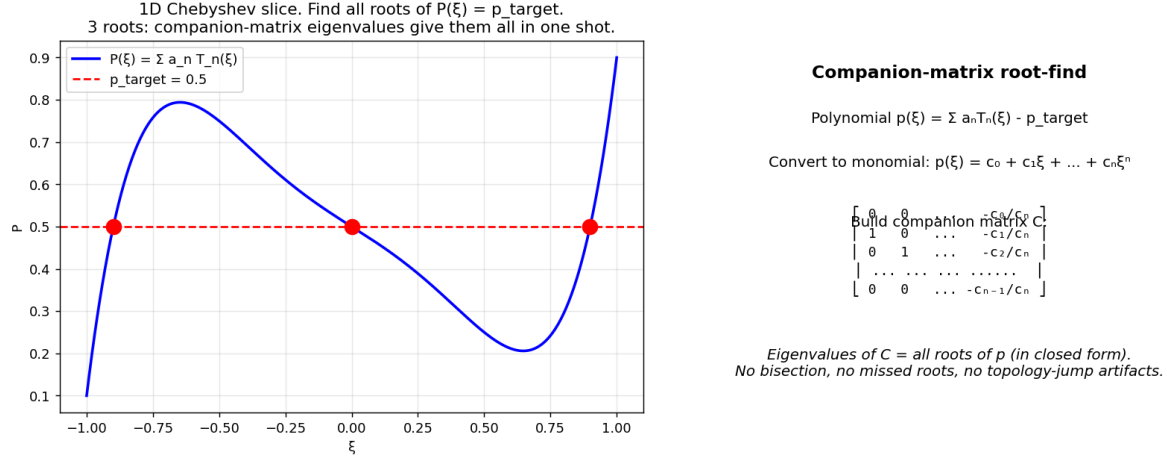


Figure 13: Companion-matrix root-finding. Left: a 1-D Chebyshev slice of P with multiple roots of $P = p_{\text{target}}$. Right: the companion-matrix construction. Eigenvalues of C give all roots in closed form (one eigendecomposition, $O(N^3)$ work) — no bisection, no missed roots, no topology-jump artifacts when the polynomial is near-critical.

This is the second big improvement Chebyshev brings over splines: the fundamental geometric primitive (contour root-find) becomes *exact*. The Morse-critical-price problem of strict- $h=0$ — where two roots merge or are born and the spline root-finder misses or doubles them — is replaced by a numerical eigenvalue problem whose discriminant detects the near-coincidence analytically.

8 Putting it all together: storage, evaluation, iteration

Final shape of the discretization:

1. **Storage:** one scalar α and one array of $\sim N^3/12$ real coefficients b_{ijk} for h , indexed by $(i \leq j \leq k, i + j + k \text{ odd})$.
2. **Evaluation of $P(u_1, u_2, u_3)$ at any point:** apply $\tanh$ to get ξ , evaluate $h(\xi)$ via the symmetric basis S_{ijk} , evaluate $T = \tau \sum u_k$, return $\sigma(\alpha T + h(\xi))$. For agent k 's slice, permute the inputs $(u_k, u_a, u_b) \rightarrow (u_{\sigma(1)}, u_{\sigma(2)}, u_{\sigma(3)})$ — S_3 invariance guarantees the same value.
3. **Operator Φ per cell:** collapse the 3-D Chebyshev polynomial to a 1-D Chebyshev polynomial in the contour direction (fix two coordinates), find all roots via the companion matrix, evaluate ∇P via Chebyshev derivative formula, integrate the co-area density via Gauss-Legendre on the fundamental domain.
4. **Iteration:** NK / Anderson on the small reduced array (α, b) . Jacobian is dense $\sim N^3/12$ by $N^3/12$ — at $N = 12$ that's $\sim 228 \times 228$, dense Newton is cheap.
5. **Refinement $N \rightarrow N + 1$:** append zero coefficients for the new modes. The old solution is an **exact** warm-start (no interpolation, no noise injection). This is the property you originally asked about that grids don't have.
6. **γ -continuation:** small step in γ ; warm-start (α, b) from the previous γ ; NK from this is typically very fast (2–5 iters at strict tolerance).

9 Summary of gains over the cube-grid / spline approach

	cube grid + spline (prior)	Chebyshev tensor + $S_3 \times Z_2$ + sigmoid lift
unknowns at " $G = 12$ equiv."	2197	$\sim 228 (228 + 1)$
refinement warm-start	interpolated, lossy	exact (append zeros)
boundary handling	clip/pin, manual	exact (sigmoid asymptote)
root-finder	bisection on spline	companion matrix (exact, all roots)
Morse-critical noise	1e-3 floor	detectable + handleable
γ -continuation	spline interp on each G	1-D in α , smooth
per-agent slice access	oblique re-interp	free (permute inputs)
asymmetric numerical noise	$\sim 1e-3$ in residual	0 (by basis)

The combination of (i) Chebyshev for spectral accuracy and exact root-find, (ii) $S_3 \times Z_2$ symmetry for reduced DOFs and free per-agent access, and (iii) the sigmoid lift for exact boundaries, addresses every limitation the prior session's strict- $h=0$ work hit. The architectural fit with MIZN is clean: this is one new file in **step2_learning/** that slots into the existing dispatcher; no other step changes.